

AI Agents and Agentic AI with Python & Generative AI

Module 1

Agentic AI Concepts

- ✓ Introduction
Video • 10 min
- ✓ Flipped Interaction Pattern
Video • 7 min
- ✓ The Agent Loop
Video • 8 min
- ✓ Knowledge Check: Agent Loop & Flipped Interaction
Practice Assignment • Grade: 100%
- ✓ Running the Code Samples in the Course
Reading • 10 min
- ✓ Programmatic Prompting for Agents
Ungraded Plugin • 15 min
- ✓ Try Out Programmatic Prompting
Reading • 10 min
- ✓ Programmatic Prompting for Agents II
Ungraded Plugin • 15 min
- ✓ Programmatic Prompting for Agents III
Ungraded Plugin • 15 min
- Programmatic Prompting: Building Blocks for AI Agents
Dialogue • 15 min
- ✓ Try Out the Customer Service Agent
Reading • 10 min
- ✓ Giving Agents Memory
Ungraded Plugin • 15 min
- ✓ Practicing Programmatic Prompting for Agents
Ungraded Plugin • 15 min
- Practicing Programmatic Prompting for Agents (Solution)
Ungraded Plugin • 15 min
- ✓ Adding Structure to AI Agent Outputs
Video • 14 min
- ✓ Building Your First Agent
Ungraded Plugin • 15 min
- ✓ AI Agent / Environment Interface
Ungraded Plugin • 15 min
- ✓ AI Agent Feedback and Memory
Ungraded Plugin • 15 min
- The Agent Loop: From Understanding to Thinking Like an Agent
Dialogue • 15 min
- ✓ Learning More & Staying Connected
Reading • 10 min
- ✓ Understanding Agentic AI Concepts
Graded Assignment • Grade: 100%

Module 2

AI Agents, Tools, Actions, & Language

- ✓ GAIL - Goals, Actions, Information, Language
Video • 7 min
- ✓ Giving Agents Tools
Video • 7 min
- ✓ Tool Descriptions and Naming
Video • 9 min

Try Out an Agent that Calls Python Functions

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2x



Try out this agent that can call Python functions.

When you run the second block in the notebook, the agent will prompt you for what action to take. You can say something like "tell me the files in the current directory" and it will make the appropriate tool choice. Experiment with the agent to see what it can and cannot do.

Look for an input like this beneath the second code block to type your command into and hit enter. Remember, this agent can only list the files in the directory and read files.

... What would you like me to do?

You should see output that looks like this after you type a command and hit enter:

```
What would you like me to do? list the files in the current directory
Agent thinking...
Agent response: <First, I will use the "list_files" tool to list all files in the current directory. This will help to identify which files are available.>
'''action
{
  "tool_name": "list_files",
  "args": {}
}
'''
Action result: {'result': ['.config', 'sample_data']}
Agent thinking...
Agent response: <The result indicates that there are two items in the directory: ".config", and "sample_data". Since these could potentially be folders, I cannot d
'''action
{
  "tool_name": "terminate",
  "args": {
    "message": "The current directory contains two items: '.config' and 'sample_data'."
  }
}
'''
The current directory contains two items: '.config' and 'sample_data'.
```

https://colab.research.google.com/drive/1W3LEOFjAQs69PJ3rM1aYG8Cofo_de6XH?usp=sharing

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